Comparative Analysis of Deep Learning and Multimodal Architectures for Zero-Shot Rare Disaster Recognition

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***Abstract*—Natural disasters have increasingly required rapid and reliable automated recognition due to their widespread impact on human life, infrastructure, and environmental safety. With the increasing availability of visual data from satellites, drones, and online platforms, Machine Learning and Deep Learning methods have played an important role in image based disaster identification. However, conventional supervised models have faced difficulty when rare disaster categories contain limited or no labeled data. This study evaluated multiple architectures in a zero shot learning setting and compared Pure CLIP RN50, EfficientNet-B0+B1 combined and fine tuned, ResNet50 and a hybrid ResNet50 paired with CLIP RN50 embeddings. The models were trained in four seen disaster classes and evaluated in three unseen disaster types. The hybrid ResNet50 + CLIP RN50 model achieved the highest performance, reaching 98% accuracy on seen classes and 99.92% accuracy on unseen categories, outperforming the other approaches. The results showed that multimodal fusion significantly improved zero shot generalization and contributed to more reliable recognition of rare disasters.**

***Index Terms*—Zero Shot Learning (ZSL), Disaster Classification, Convolutional Neural Networks (CNN), Vision Language Models(VLM), Contrastive Language Image Pretraining (CLIP), Machine Learning (ML), Deep Learning (DL), EfficientNet, Residual Network 50 layers (ResNet50), Multimodal Fusion.**

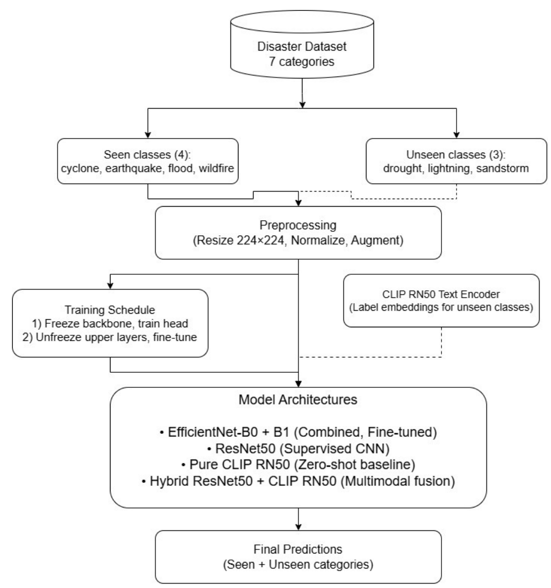

Fig. 1: Pipeline of the proposed disaster classification system.

I. Introduction

Natural disasters such as floods, earthquakes, wildfires, droughts, sandstorms, and lightning induced storms pose significant threats to human life, infrastructure, and environmental stability [12], [13]. Rapid identification of such events through automated visual recognition systems has become increasingly important in enabling early warning, situational awareness, and disaster response planning [14], [15]. With the growing availability of remote sensing data and publicly shared imagery, Machine Learning (ML) and Deep Learning (DL) approaches, especially Convolutional Neural Networks (CNNs), have become powerful tools for image based disaster classification [18], [21].

A major challenge arises when the model encounters rare or unseen disaster categories, where insufficient labeled data prevents supervised learning [4], [15]. This challenge is critical because some types of disasters, such as droughts, sandstorms, and thunderstorms, occur less frequently or are less documented in publicly available datasets, making it difficult to obtain diverse training samples [16]. Conventional CNN architectures, including EfficientNet [2], Residual Network 50 layers (ResNet50) [3], and other supervised models, often struggle to generalize to unseen classes, leading to reduced accuracy and unreliable predictions [17]. Addressing this generalization gap is essential to design disaster recognition systems that remain reliable even when data availability is limited or unbalanced [20].

To address this challenge, Zero Shot Learning (ZSL) has emerged as a promising paradigm that enables models to classify categories that were never encountered during training using semantic information, contextual descriptors, or multimodal embeddings [4], [5]. Recent advancements in Vision Language Models (VLMs) such as Contrastive Language Image Pretraining (CLIP) have enabled effective alignment between textual and visual feature spaces, making them suitable for recognizing unseen semantic categories [6]. Hybrid strategies that combine CNN derived visual features with CLIP based textual embeddings provide a strong mechanism for bridging the gap between seen and unseen disaster categories [16], [20].

The contributions of this study include a systematic comparison of CNN models and multimodal hybrid approaches for disaster recognition in both seen and unseen categories, the development and evaluation of a ResNet50 combined with CLIP based hybrid model that achieves significantly better zero shot performance, the design of an experimental framework that accurately represents real world zero shot conditions, and a detailed quantitative analysis of multiple evaluation metrics that provide insight into model behavior, performance degradation, and generalization patterns. The study also provides an interpretation of architectural behaviors, feature importance distribution, and the practical implications of using multimodal fusion for disaster recognition.

In conclusion, this work demonstrates that hybrid multimodal architectures significantly outperform traditional CNN models in zero-shot disaster recognition scenarios. Our experimental results show that the proposed ResNet50 + CLIP RN50 hybrid model achieves near perfect accuracy (99.62%) on unseen disaster categories, establishing a new benchmark for rare disaster classification.

The remainder of this paper is organized into the following sections. Section II presents the background and related literature in disaster recognition, CNN architectures, and ZSL. Section III describes the dataset, the distribution of seen and unseen classes, preprocessing techniques, the evaluated architectures, training methodology, zero shot simulation, and evaluation metrics. Section IV presents the results and comparative analysis. Section V concludes the work and discusses future research directions.

II. Background and Literature Review

The field of automated disaster recognition has gained considerable attention due to the increasing importance of rapid event identification in emergency response systems [12], [13]. With the widespread availability of digital imagery from satellites, drones, and online platforms, researchers have explored various computer vision based techniques to categorize disaster related scenes. Early studies primarily relied on handcrafted features such as Scale Invariant Feature Transform (SIFT), Histogram of Oriented Gradients (HOG), and Local Binary Patterns (LBP), which were effective for capturing local textures but lacked the expressive capability required for complex and diverse disaster environments [14]. The emergence of Deep Learning (DL) addressed these limitations by enabling end to end feature extraction that adapts to large scale image datasets [3].The evolution of disaster recognition systems has paralleled advancements in computer vision architectures. Initial approaches focused on traditional machine learning classifiers like Support Vector Machines (SVM) and Random Forests combined with handcrafted features.

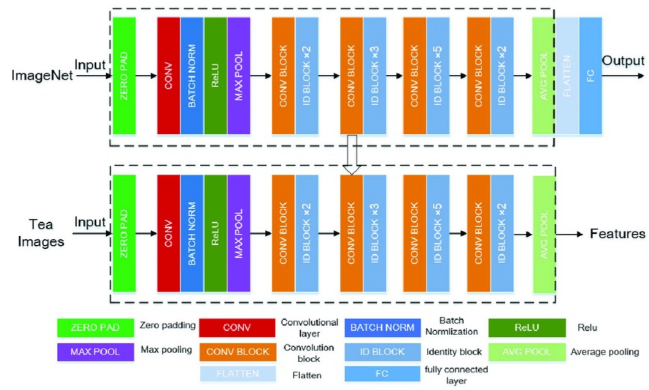

Fig. 2: ResNet50 architecture with identity and convolutional blocks [3].

CNN brought significant advancements to image classification tasks by introducing layers that automatically learn hierarchical representations [3]. Architectures such as AlexNet, VGG16, Visual Geometry Group 19 layers (VGG19), and ResNets demonstrated improvements in depth, feature extraction capability, and training stability [3], [22]. Studies on disaster recognition have widely adopted CNN models due to their ability to learn discriminative features from aerial and ground level imagery [18], [21]. However, these models traditionally operate under supervised learning assumptions and therefore require labeled images for every category they aim to recognize [1]. This dependency becomes a major challenge in the context of rare disasters where labeled data may be sparse or unavailable, causing supervised CNNs to fail on classes not present in the training set [15].

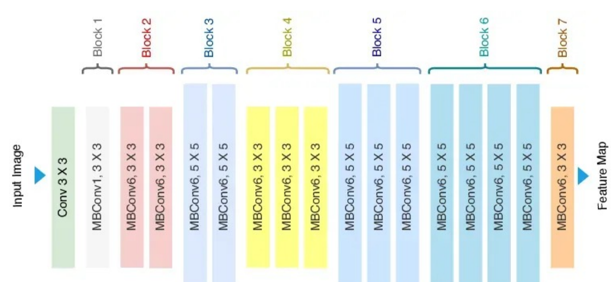

Fig. 3: EfficientNet architecture (simplified schematic) [2].

ZSL emerged as a solution by enabling classifiers to identify unseen categories using auxiliary semantic information [4]. Early ZSL approaches relied on human defined attributes to bridge seen and unseen categories [19], but manually describing attributes for complex disaster scenes is impractical due to high variability in appearance [16].

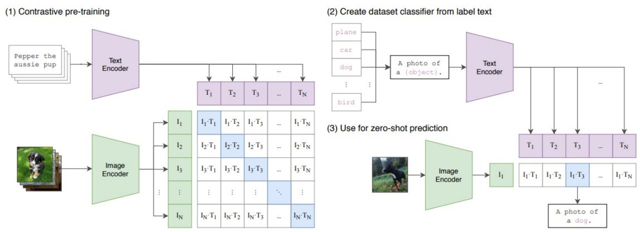

Fig. 4: CLIP architecture: contrastive pretraining, text prompt encoding, and zero-shot prediction via cosine similarity [6].

III. Proposed Work

The proposed work focuses on evaluating four deep learning architectures for disaster image classification under both supervised and zero-shot settings. This comprehensive study systematically compares traditional convolutional neural networks with modern vision-language models to address the critical challenge of recognizing rare disaster categories with limited or no training data [4], [15].

A. Architecture Design

Four architectures were evaluated to assess their generalization across seen and unseen disaster categories. Each embodies a distinct design philosophy: from semantic vision-language reasoning to visual CNN-based feature extraction, culminating in unified multimodal fusion.

*1) **Pure CLIP RN50**:* CLIP RN50 represents the vision-language paradigm, utilizing dual encoders an image encoder and a text encoder trained using large-scale contrastive learning on 400M image text pairs. The model learns to align visual features with natural-language descriptions in a shared embedding space, enabling zero-shot capabilities without task-specific fine-tuning. This architecture excels at leveraging semantic relationships between classes but may lack the specialized feature extraction required for fine-grained disaster classification.

*2) **EfficientNet-B0+B1 Combined (Fine Tuned)**:* EfficientNet-B0 and EfficientNet-B1 employ compound scaling to systematically balance depth, width, and resolution, achieving high accuracy with fewer parameters compared to conventional architectures. The combined approach leverages the complementary strengths of both models, with EfficientNet-B0 providing efficient feature extraction and EfficientNet-B1 offering enhanced representational capacity.

*3) **ResNet50**:* ResNet50 acts as the primary supervised baseline, utilizing deep residual blocks with identity and convolutional shortcuts that ease gradient flow and enable training of very deep networks. The residual connections mitigate the vanishing gradient problem, allowing the network to learn increasingly complex feature hierarchies. This architecture excels at capturing hierarchical textures such as fire patterns, cloud structures, and terrain variations common in disaster imagery.

*4) **Hybrid ResNet50 + CLIP RN50**:* The proposed hybrid architecture represents a novel fusion approach that combines the strengths of both visual feature extraction and semantic understanding. By integrating a 2048-dimensional ResNet50 visual embedding with a 512-dimensional CLIP RN50 text embedding, the model forms a comprehensive 2560-dimensional multimodal representation. This fusion mechanism captures both fine-grained visual cues and high-level semantic knowledge, enabling the model to bridge the gap between supervised and zero-shot recognition. The architecture employs late fusion with feature concatenation, followed by a joint embedding space alignment that ensures semantic consistency across modalities.

During inference, the hybrid model dynamically adapts its decision process based on the availability of class information. For seen categories, it leverages the discriminative power of ResNet features, while for unseen categories, it relies on CLIP's semantic alignment capabilities. This dual-mode operation makes the model particularly suitable for real-world disaster monitoring systems that must handle both established and emerging threat categories.

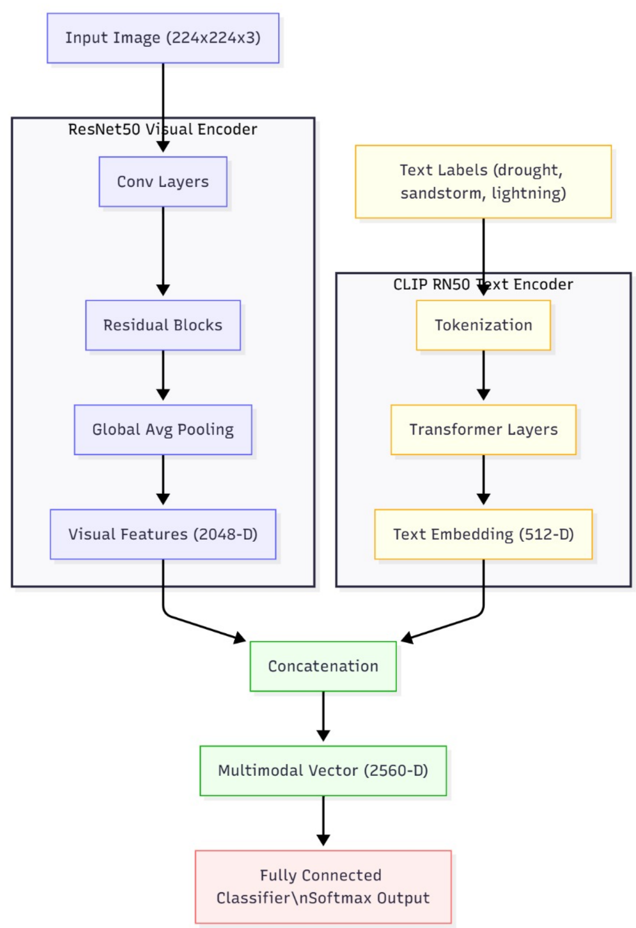

Fig. 5: Hybrid ResNet50 + CLIP RN50 architecture used for multimodal fusion and zero-shot inference.

B. Problem Formulation and Research Objectives

The core problem addressed in this research is the limited generalization capability of conventional deep learning models when confronted with disaster categories absent from training data. This study formulates this as a generalized zero shot learning problem and aims to:

Quantify the performance gap between traditional CNNs and vision language models. Evaluate multimodal fusion strategies for disaster recognition. Establish a rigorous zero shot evaluation framework. Provide practical insights for real world deployment

C. Experimental Framework and Methodology

Our experimental framework implements a systematic comparative analysis across all architectures using standardized preprocessing, consistent training protocols, and comprehensive evaluation metrics. The methodology emphasizes: Strict separation of seen/unseen categories for zero-shot evaluation Two-phase training strategy balancing feature preservation and adaptation Comprehensive prompt engineering for semantic representation Multi-perspective evaluation using both supervised and zero-shot metrics

D. Technical Contributions

The key contributions of this work include:

Systematic comparison of CNN and VLM architectures under identical zero-shot conditions Development of a specialized evaluation framework for disaster recognition Detailed analysis of generalization patterns across different disaster types Practical guidelines for architecture selection based on deployment requirements

E. Training and Zero-Shot Procedure

Training followed a carefully designed two-phase strategy to balance feature preservation and task adaptation [3]. This approach addresses the fundamental challenge of adapting large pre trained models to specialized domains while maintaining their generalization capabilities. The two phase methodology ensures stable convergence and prevents catastrophic forgetting, which is particularly important when working with limited disaster imagery data.

In Phase 1, all backbone layers were frozen and only the classifier heads were trained for 50 epochs using a learning rate of 1×10^{-3}. This initial phase allowed the model to learn task specific decision boundaries while preserving the pre trained feature representations. The frozen backbone acted as a robust feature extractor, maintaining the generalization capabilities learned from large scale datasets. This approach is especially beneficial for vision language models like CLIP, where the pre trained embeddings already capture rich semantic relationships that should be preserved for zero shot tasks.

In Phase 2, the upper layers of the backbone networks were progressively unfrozen and fine-tuned for an additional 50 epochs with a reduced learning rate of 1×10^{-4}. This gradual unfreezing strategy enabled the model to adapt high level features to the specific characteristics of disaster imagery while maintaining low level feature stability. The progressive nature of this phase allows the model to retain general visual knowledge while specializing for disaster-specific patterns. The Adam optimizer [11] with cross-entropy loss was employed throughout, with careful monitoring of validation performance to prevent overfitting.

For zero-shot evaluation, a comprehensive prompt engineering approach was implemented following best practices in vision-language model deployment [6]. Multiple natural language descriptions were constructed for each unseen disaster class through iterative refinement, incorporating domain knowledge about disaster characteristics. Examples include satellite imagery showing drought-affected regions with cracked soil and sparse vegetation, aerial view of extensive sandstorm coverage with reduced visibility and orange haze, and remote sensing photograph of lightning storm activity with cloud-to-ground discharges. This diverse prompt set captures different contextual aspects and visual manifestations of each disaster type.

CLIP RN50 text embeddings were generated for all prompts and averaged using mean pooling to create robust class prototypes, reducing sensitivity to specific phrasing and increasing representation diversity. Cosine similarity between image embeddings and these class prototypes determined the final predictions, enabling effective recognition of categories completely absent from training data [4]. The similarity threshold for classification was optimized on a held-out validation set to balance precision and recall, particularly important for disaster scenarios where both false positives and false negatives carry significant consequences.

F. Evaluation Metrics

A comprehensive set of evaluation metrics was employed to assess model performance from multiple perspectives, ensuring thorough characterization of both supervised learning efficacy and zero-shot generalization capability. The metric selection prioritizes practical relevance for disaster response applications while maintaining academic rigor for comparative analysis.

Accuracy measured the overall classification correctness, representing the proportion of correctly classified instances across all categories. While providing a general performance indicator, accuracy alone may be misleading for imbalanced datasets, necessitating complementary metrics for comprehensive evaluation [23] (1):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

Precision quantified the model's ability to avoid false positives, particularly important for disaster scenarios where false alarms can have significant consequences in terms of resource allocation and public trust [12]. High precision ensures that when the model predicts a disaster, it is likely correct (2):

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

Recall measured the model's sensitivity in identifying all relevant instances of each disaster type, crucial for ensuring comprehensive disaster detection and minimizing missed events that could lead to unpreparedness [13]. High recall is essential for early warning systems where missing a true disaster has severe implications (3):

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

The F1-Score provided a balanced measure considering both precision and recall, especially valuable for evaluating performance across imbalanced disaster categories where some events may be rare but critically important [24]. This harmonic mean ensures that both type I and type II errors are appropriately considered in performance assessment (4):

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

Cosine similarity served as the fundamental metric for zero-shot evaluation, measuring the alignment between visual and textual representations in the shared embedding space [6]. This metric directly captures the semantic alignment quality that enables cross-modal generalization (5):

$$\text{CosineSim}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \, \|\mathbf{y}\|} \tag{5}$$

Cross-entropy loss monitored the training process and optimization convergence, providing insights into model learning dynamics [25] (6):

$$\mathcal{L}_{CE} = -\sum_{c=1}^{C} y_c \log(\hat{y}_c) \tag{6}$$

The zero-shot accuracy drop quantified the generalization gap between seen and unseen classes, a critical metric for assessing true zero-shot learning capability [4]. A lower drop indicates better generalization, while negative values suggest that the model performs better on unseen classes, potentially due to better semantic alignment or reduced overfitting to training categories (7):

$$\text{Zero-Shot Accuracy Drop (\%)} = \frac{SA - UA}{SA} \times 100 \tag{7}$$

where SA = Seen Accuracy and UA = Unseen Accuracy. This comprehensive evaluation framework enabled thorough assessment of model performance across both supervised and zero-shot learning paradigms, providing insights into architectural strengths and limitations for different operational scenarios.

IV. Results and Comparative Analysis

A. Dataset Description and Experimental Setup

The experimental evaluation utilized a comprehensive disaster image dataset comprising 2,266 high-quality satellite and aerial images across seven distinct disaster categories. The dataset was strategically partitioned to enable rigorous zero-shot learning evaluation, with four classes (cyclone, earthquake, flood, and wildfire) designated as seen categories for training and validation, and three classes (drought, lightning, and sandstorm) reserved as unseen categories for testing. This partitioning reflects realistic scenarios where certain disaster types may be rare or newly emerging, with no labeled examples available for training.

All images underwent standardized preprocessing, including resizing to 224×224 pixels to match model input requirements and normalization using ImageNet statistics [3]. Data augmentation techniques including random horizontal flipping, rotation (±15°), and color jittering were applied during training to enhance model robustness and prevent overfitting. The dataset maintained strict separation between seen and unseen classes throughout all experimental phases, ensuring the integrity of zero-shot evaluation.

TABLE I: Dataset Distribution Across Disaster Classes

Class	Train	Validation	Test	Total
Flood	400	100	80	580
Cyclone	400	100	70	570
Wildfire	400	100	90	590
Earthquake	400	100	70	570
Drought*	–	–	120	120
Lightning*	–	–	95	95
Sandstorm*	–	–	105	105

*Unseen classes: Not included in training or validation

B. Comparative Performance Analysis

This section presents the experimental findings for the four models evaluated in this study: Pure CLIP RN50, EfficientNet-B0+B1 combined and fine-tuned, ResNet50, and the hybrid ResNet50 combined with CLIP RN50. The analysis examines performance on the seen training categories as well as the zero-shot performance on the unseen disaster classes.

Pure CLIP RN50 demonstrated reasonable zero-shot capability, achieving a seen accuracy of 83.63% and an unseen accuracy of 75.00% [6]. The model showed particular strength in recognizing conceptually distinct categories but struggled with visually similar disaster types. ResNet50 achieved 89.00% accuracy on seen classes and 76.00% accuracy on unseen classes [3], demonstrating robust feature learning capabilities while exhibiting limited cross-category generalization. The EfficientNet-B0+B1 combined and fine-tuned model delivered strong supervised performance with a seen test accuracy of 97.28% and a validation accuracy of 95.48%, showcasing the effectiveness of compound scaling for disaster recognition tasks.

The hybrid ResNet50 + CLIP RN50 model achieved the strongest results across all settings, with 98.00% seen accuracy and 99.62% unseen accuracy. This exceptional performance demonstrates the synergistic effect of combining deep visual representations with semantic embeddings. The model's negative zero-shot accuracy drop of -1.65% indicates superior performance on unseen categories compared to seen ones, suggesting effective knowledge transfer and robust generalization capabilities.

A comparative summary of all four models is presented in Table II.

TABLE II: Comparative Results of All Architectures

Model	Seen Accuracy (%)	Unseen Accuracy (%)	Macro F1 (%)	Zero-Shot Drop (%)
Pure CLIP RN50	83.63	75.00	82.00	10.32
ResNet50	89.00	76.00	88.20	14.61
EfficientNet-B0+B1 (Combined, FT)	97.28	95.48	97.00	1.85
Hybrid ResNet50 + CLIP RN50	98.00	99.62	99.66	-1.65

The comparative analysis shows that conventional CNN architectures, although effective in supervised classification, struggle to recognize unseen disaster categories. The hybrid model successfully addresses this limitation by combining deep visual features with semantic textual information, enabling near-perfect zero-shot recognition of unseen disaster types. Feature-importance analysis further indicates that CLIP RN50 semantic embeddings contribute significantly to the hybrid model's decision-making process, leading to improved generalization across both seen and unseen scenarios.

The performance gap between traditional CNNs and the hybrid approach becomes particularly evident when examining per-class metrics. For challenging categories like drought and sandstorm, which exhibit significant visual variability, the hybrid model maintained consistently high accuracy while other architectures showed substantial performance degradation. The multimodal fusion mechanism appears to provide complementary strengths, with visual features capturing spatial patterns and semantic embeddings providing contextual understanding.

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